

Emilien Breton

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— EXPERIENCE —

- EcoSafeSense**
FIRMWARE ENGINEER

Ottawa | October 2024–Present

 - Writing firmware for an ESP32-based air quality sensor and complementary test bench.
- Cohere**
DATA QUALITY SPECIALIST — ADVANCED MATHEMATICS

Freelance | October 2024–April 2025

 - Wrote, audited and corrected LLM prompts and responses to produce spotless training data in formal logic, combinatorics, number theory, graph theory and mathematical optimization.

— PROJECTS —

- DFA Regex Engine**
 - Built a regex engine in C99 that compiles regular expressions down to minimal deterministic finite automata to match input strings in linear time without backtracking.
 - Wrote a test suite of over 500 end-to-end tests to ensure correctness of the engine and catch regressions.
 - Developed a grep-like tool to stress-test the engine and achieved performance on par with GNU grep.
- Multilayer Perceptron**
 - Wrote a static reverse-mode automatic differentiation library in C99 for use in deep learning.
 - Designed a multilayer perceptron model atop the library and implemented stochastic gradient descent with momentum, ultimately achieving 97% accuracy on the MNIST database after 3 minutes training on 16 threads.
- Purely Functional Language**
 - Developed an interpreter for a lazily-evaluated untyped λ -calculus augmented with monadic I/O.
 - Bootstrapped the interpreter into a usable language by writing a 2500-line prelude complete with booleans, integers, pairs, optionals, lists, strings, and type classes for total orders, monoids, functors, monads, foldables...
- Breadboard Microcomputer**
 - Designed an 8-bit microcomputer from logic gates upwards, including an assembler, emulator and rudimentary C compiler in Rust, totaling over 20 000 SLOC and 750 hours of work.
 - Wrote various programs running natively on the microcomputer, including a Wozmon-like memory monitor, 16×16 sprite editor, native assembler, postfix notation calculator and Tetris clone.
 - Built the microcomputer in hardware by hand-wiring discrete 74HC-series logic chips on breadboards.

— VOLUNTEERING —

- Computer Science Club**
CLUB EXECUTIVE

University of Ottawa | June 2022–Present

 - Running a growing community of over 1500 computer science students at the University of Ottawa.
 - Collaborating with the executive board to brainstorm, plan, fund and market monthly events and meetups, such as workshops on Vim bindings and a mini-course on the λ -calculus.
- Hack the Hill Hackathon**
DEVELOPMENT MANAGER — DEVELOPMENT TEAM

Ottawa | November 2022–October 2024

 - Led development of an open-source event management system based on Next.js and Prisma and used by over 1000 hackers and 50 organizers throughout the hackathon.
 - Built and maintained an internal payment portal powered by Stripe and React.js in collaboration with the sponsorship team that processed upwards of 20 000\$.

— AWARDS —

- IEEE x HTB CTF — 2nd place
 - Tech-NoI-Hack 2025 — 2nd place
 - AIT Hackathon 3.0 — 1st place, Mocha track
 - uOCTF 2025 — 1st place
 - uOttHack 6 — 1st place, QNX track
 - CS Games 2024 — 1st place, IoT track
 - uOCTF 2024 — 1st place
 - DeFi The Conventional — 1st place, DeFi track
- January 2026

October 2025

September 2025

March 2025

January 2025

March 2024

November 2023

March 2022

— SKILLS —

- Languages — C • Rust • Python • Haskell • JavaScript
 - Tools — GNU/Linux • GDB • GNU Make • Vim • React
- f4f411e5 • May 2026